Key findings from "Crowdsourced Data: Accuracy, Accessibility, and Authority (CDAAA)"

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In a systematic literature review on the topic of accessibility of DH projects and digital cultural heritage for people with disabilities, in the period 2018 to 2022, Pirrone et al (2023) found too few sources to answer their initial question "RQ1: How Digital Humanities are supporting people with impairment in cultural enjoyment?". The dearth of scholarship on the topic of accessibility and DH caused them to change their question to be future-oriented and speculative: "RQ2: How Digital Humanities might improve the quality of the cultural experience for people with disabilities?" (Pirrone et al, 2023, Table 1, emphasis added). The paucity of sources is particularly notable given that their study considered all forms of disability. Given these findings, the DH25 conference theme is timely.

The research questions and findings from "Crowdsourced Data: Accuracy, Accessibility, and Authority (CDAAA)," a 3-year Institute of Museum and Library Services (IMLS, USA) grant project offers new insights into questions of accessibility of the cultural heritage record. CDAAA aimed to identify any barriers that Library, Archive, and Museum (LAM) professionals and academics face in making crowdsourced transcriptions accessible to blind and low vision (BLV) people who use assistive technology to access digital text; as well as solutions to these issues. Between 2022 and 2025, the CDAAA team, led by Victoria Van Hyning, set out to understand whether the crowdsourced transcription projects deployed by LAM organizations have worked as many

creators of such projects believed they would.

Our research questions are: RQ1 (Authority): Are LAMs able to integrate crowdsourced transcriptions into their CMSs (the authoritative record)? If yes, how? If not, what technical barriers do they face? RQ2 (Accuracy and Authority): What are LAM practitioners' attitudes towards crowdsourced transcription data quality? Do these attitudes impact whether or not crowdsourced data are incorporated into CMSs? How do LAM practitioners assess the quality of crowdsourced data? RQ3 (Accessibility): When transcription data is successfully integrated with CMSs, is it accessible to print-disabled people? What are print-disabled users' experiences of searching for and reading transcription data?

To answer these questions, we worked with two key constituencies: project owners and users of the resulting data:

12 US-based Partners, including representatives from private, public, university, and federal LAM institutions, and one community-led group, who use platforms like Zooniverse.org, FromThePage, and/or bespoke software to run transcription projects. We selected LAM Partners who also have responsibility for managing the resulting crowdsourced transcriptions and any other data, i.e. tags or index terms.

11 blind and 1 low vision users with diverse academic backgrounds and training.

Methods

We used a mixed-methods approach, including a recruitment survey, semi-structured interviews, and technical demonstration interviews with LAM partners; and accessibility focused user-testing sessions with BLV participants. LAM Partners were recruited through a survey sent to multiple mailing lists, posted on LinkedIn, and shared by members of the Advisory Board. LAM Partner interviews were conducted between 2023-06-28 and 2024-04-15, and resulted in 21 recordings, including a pilot interview. The 3 LAM

Partners who had not yet integrated transcription data into a LAM system or data repository did not participate in a technical demonstration.

User testers were recruited through a blind academics listserv managed by the National Federation of the Blind in Baltimore, USA. The NFB vetted our research proposal before sharing the recruitment email. We recruited an exceptionally international, multilingual, and diverse user tester pool of 12 people. Each participant tested 3 LAM Partner systems in a roughly 120-minute-long session. 10 BLV participants tested an additional accessible semantic HTML rendering of a transcription of a page from the diary of Ada Blackjack held at Dartmouth College Libraries, which we presented at DH24 (Van Hyning et al, 2024).

All recordings were transcribed through Otter.ai and then manually checked by members of the CDAAA team and sent to participants who had the opportunity to amend and redact the transcript. An initial codebook for the accessibility testing was created by PI Van Hyning, Research Scientist J. Bern Jordan (Y2 of the grant only), and Graduate Assistant Mace Jones in summer 2024. We met together and read through transcripts to develop codes through discussion and consensus. The codes include in vivo, deductive and inductive types. Jones and Van Hyning separately began coding the LAM Partner transcripts in spring 2025 using Evaluative and Magnitude coding approaches (Saldaña, 2021).

Interviews, Demos, and Accessibility testing protocols all included Likert scale questions that enable us to compare findings across LAM Partner organizations and systems. For example, we asked each BLV tester to perform tasks on three LAM systems and rate their experience on a five-point scale in response to statements such as “This discovery system website is easy to use” (strongly agree, somewhat agree, neither agree nor disagree, somewhat disagree, strongly disagree). We

tested the following tasks for usability and accessibility:

Locate the website of the relevant LAM Partner

Search the website, content management system, data repository, etc for transcriptions of any kind.

Search the relevant system for a specific quotation created through the LAM Partner's transcription project and known to be housed in the platform.

These five-point scale questions are based on the System Usability Score (SUS) questionnaire that aims to make comparison between systems possible. Users were assigned LAM systems to test by a randomized approach. Each system was tested by 4 users.

LAM Partner Findings (RQ1 and RQ2)

Through Interviews and Demos, we identified that LAM Partners experience common challenges with crowdsourced data integration, including infrequent and complex manual ingest processes, however, more LAMs were able to upload content more successfully than we had anticipated (N=10/12). LAM practitioners approach data ingestion in different ways, i.e. placing transcriptions in notes or description fields, publishing a separate .txt or .pdf file attached to the object record, or as datasets in a separate repository like Dataverse.

We asked each LAM Partner participant questions about their data quality assessments and processes (if any) of vetting transcriptions. We found that 10 LAM Partners rate volunteer transcriptions as high enough quality to ingest (Table 1). Many performed some

text editing and post-processing before ingest.

Magnitude Codes	LAM A1	LAM A2	LAM A3	LAM A4	LAM A5	LAM A6	LAM A7	LAM A8	LAM A9	LAM A10	LAM A11	LAM A12
Positive			X		X		X		X (wiki)	X		
Leans positive	X (FTP)	X						X				X
Neutral	X (Zoo)										X	
Leans negative				X		X						
Negative									X (Zoo)			

Table 1: Sentiment of LAM Practitioners About Their Data Quality

LAM Partners tended to take a positive view of crowdsourced transcription quality. Some made no changes to transcriptions before ingesting them to their CMSs, while others vetted every line. The negative results require a little unpacking. LAM Partner A4's "leans negative" assessment stems from the fact that they use crowdsourcing for pedagogical purposes, and the data that results is published in Dataverse in order to provide credit and acknowledgement to students taking part in paleography institutes, rather than for the primary purpose of creating searchable and accessible transcription content. 2 LAM Partners collected crowdsourcing data that are too low quality or difficult to use and had not ingested it at the time of the interview. LAM Partner 12 had originally used the codebase Scribe, developed by Zooniverse and the New York Public Library, and finding the data quality too low, moved the content to the Zooniverse Project Builder for retranscription. LAM Partner 9 had a good experience with Wikisource, but greater challenges with the Zooniverse Project Builder. We did not find a correlation between time spent vetting transcriptions and assessments of data quality.

Accessibility User Tester Findings (RQ3)

While the data management practices of our LAM Partners often meet FAIR and CARE principles of data sharing that may work well for sighted

users, that not all data sharing methods are equal in terms of accessibility for BLV users. Depositing batches of transcriptions in repositories such as Dataverse and Zenodo–commonly used data repositories for many DH and LAM practitioners–are notably tricky, as they do not facilitate full text search, which is a significant limitation for many users, but particularly for BLV users.

As stated above, all BLV participants undertook the same series of tasks for each CMS or database. Though many located the Partner websites and often found transcriptions using a search string and iterative approaches to their search, it is notable that our participants struggled as much as they did. Many pointed out that a first encounter with a website usually requires what we call "bushwacking" and navigating through heterogeneous site architecture. Many of our participants were highly adept at this, but said that under normal circumstances they tend to abandon poorly architected websites in favaor of those that are more intuitive or better structured. It is notable that all of our participants had earned or were working towards advanced degrees and regularly conduct sophisticated online and LAM-based searches for their studies or work. The sheer variety of search architectures, website layouts, and methods for publishing crowdsourced transcription data pose challenges for BLV users, which aligns with prior findings about BLV users' search experiences in other kinds of library CMSs and websites (Xie et al, 2018 and 2021). Here is an example of findings for

just one LAM Partner's bespoke CMS:

ACC Tester ID	Blind or low vision	**Navigate to the L AM Partner website**	Search for digitized materials w/ transcriptions	Phrase search	Rate statement: 'This discovery system website is easy to use'	Rate statement: 'This website meets my needs as a user.'	LAM CMS test position
ACC-4	B	Complete Success	Failure	Failure	Strongly Disagree	Strongly Disagree	2
ACC-6	B	Complete Success	Failure	Failure	Neither Agree nor Disagree	Agree	1
ACC-8	B	Complete Success	Failure	Failure	Agree	Agree	3
ACC-11	B	Complete Success	Failure	Success w/ major issues	Agree	Agree	1

Table 2: Findings for LAM Partner A4, an R1 University Library in the USA.

Conclusion

CDAAA has many lessons for the DH and LAM community as we consider how to ensure the accessibility of our collections, scholarly editions, and other web-based content. Typically, when DH and cultural heritage practitioners say that something is accessible, we mean that it is digital and online, rather than truly accessible to people with disabilities. Our findings are especially timely, given swift-approaching changes to legislation in the USA, in the form of '35.200 Requirements for web and mobile accessibility', an update to the Americans with Disabilities Act Title II regulations, subpart-H. Effective 24 June, 2024 the Department of Justice in the USA enacted this new rule to provide "technical standards for web content and mobile app accessibility to give public entities greater clarity in exactly how to meet their ADA obligations and to help ensure equal access to government services for individuals with disabilities," which explicitly cites the WCAG 2.1 AA as the expected standard. All Title II organizations, including state and university libraries and archives, local governments, etc. must comply with these regulatory standards by 24 April of 2026 or 2027, depending on the size of the jurisdiction they serve. Similar legislation exists in Europe (European Accessibility

Act, which came into force in June 2025), and the UK (Public Sector Bodies (Websites and Mobile Applications) (No. 2) Accessibility Regulations 2018). We believe new and existing laws and statutes will impact DH and LAM work for the better and move us towards addressing many of the thematic concerns of the DH25 conference, and many of the deepest ethical concerns at the heart of our work.

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